10P

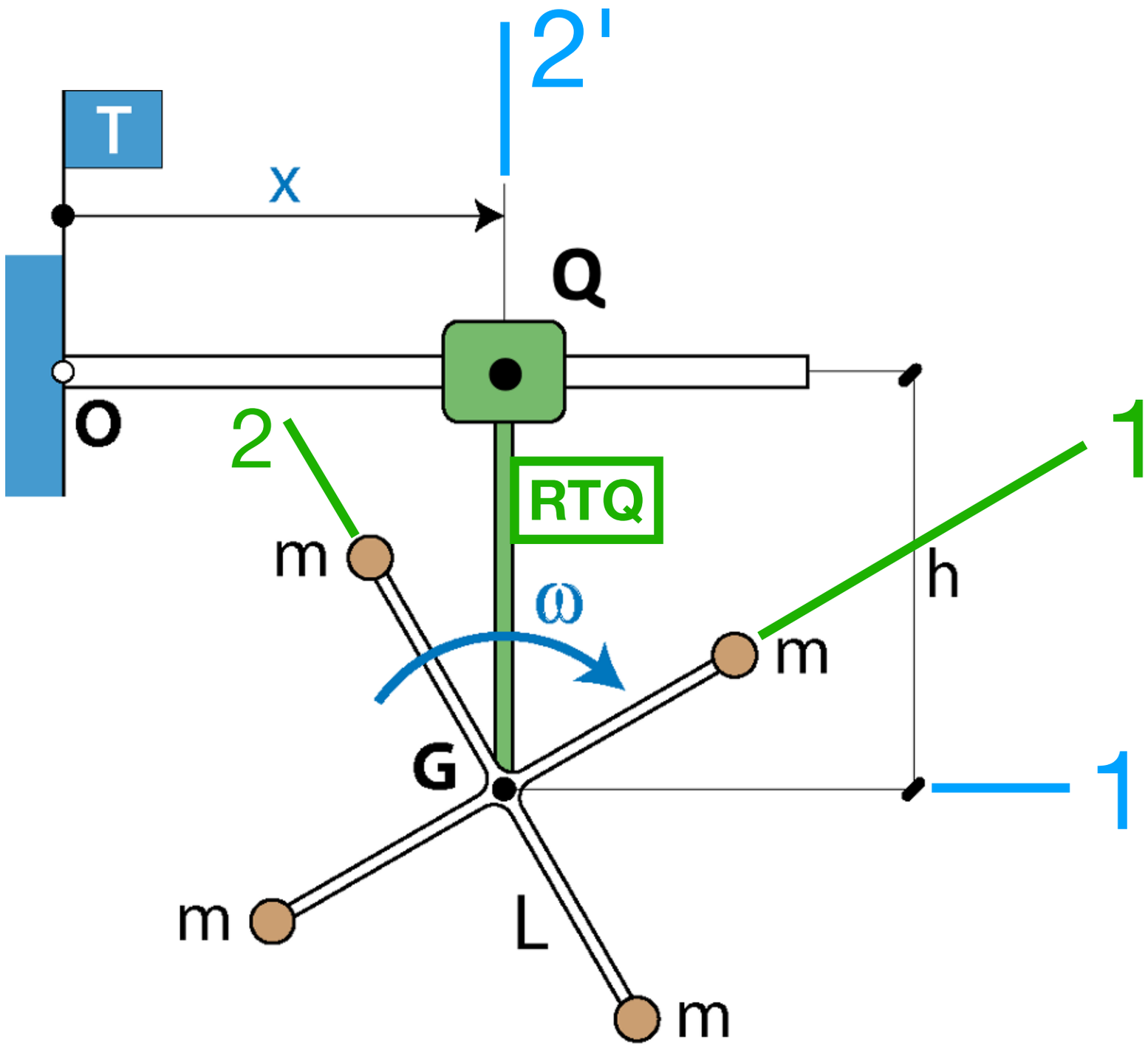
Teoremes vectorials I

Exercicis de càlcul del moment cinètic

Problema 3D

$\bar{H}_{RTQ}(\mathbf{Q})$?

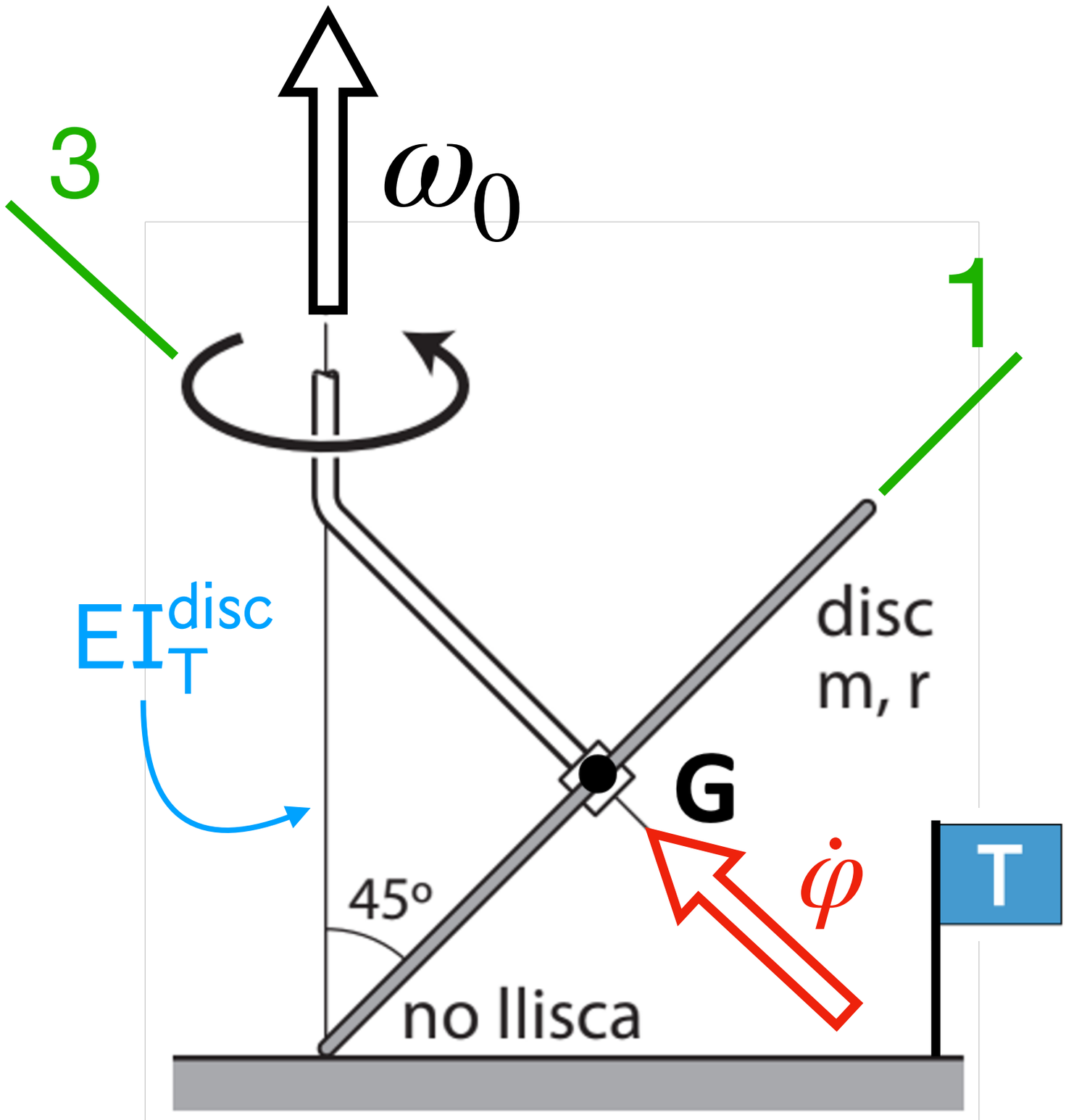
$\bar{H}_{RTO}(\mathbf{O})$?



B = (1,2,3)

B' = (1',2',3')

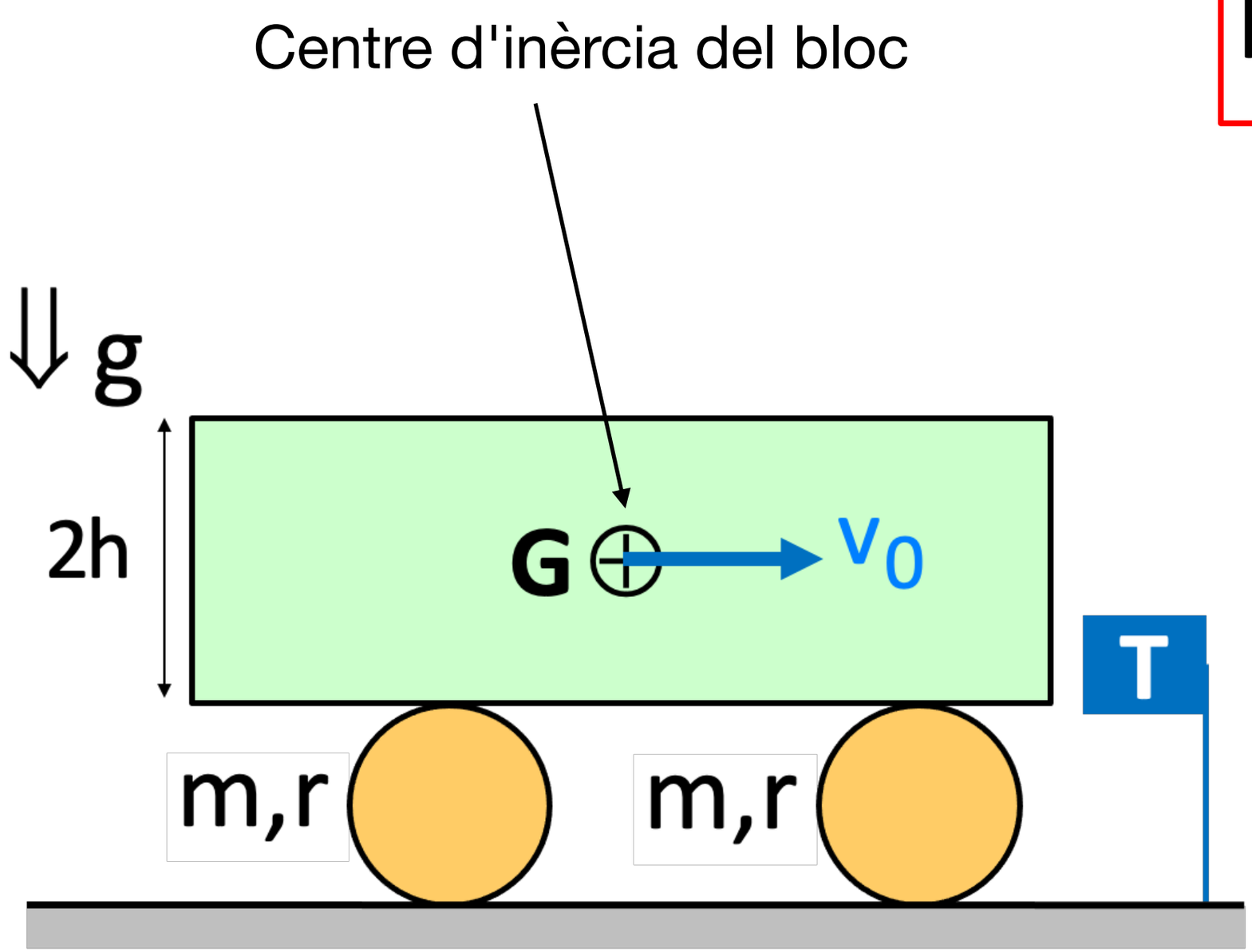
Només les 4 partícules de la creu són de massa no negligible



$$\boxed{\overline{H}_{RTG}^{disc}(\mathbf{G})?}$$

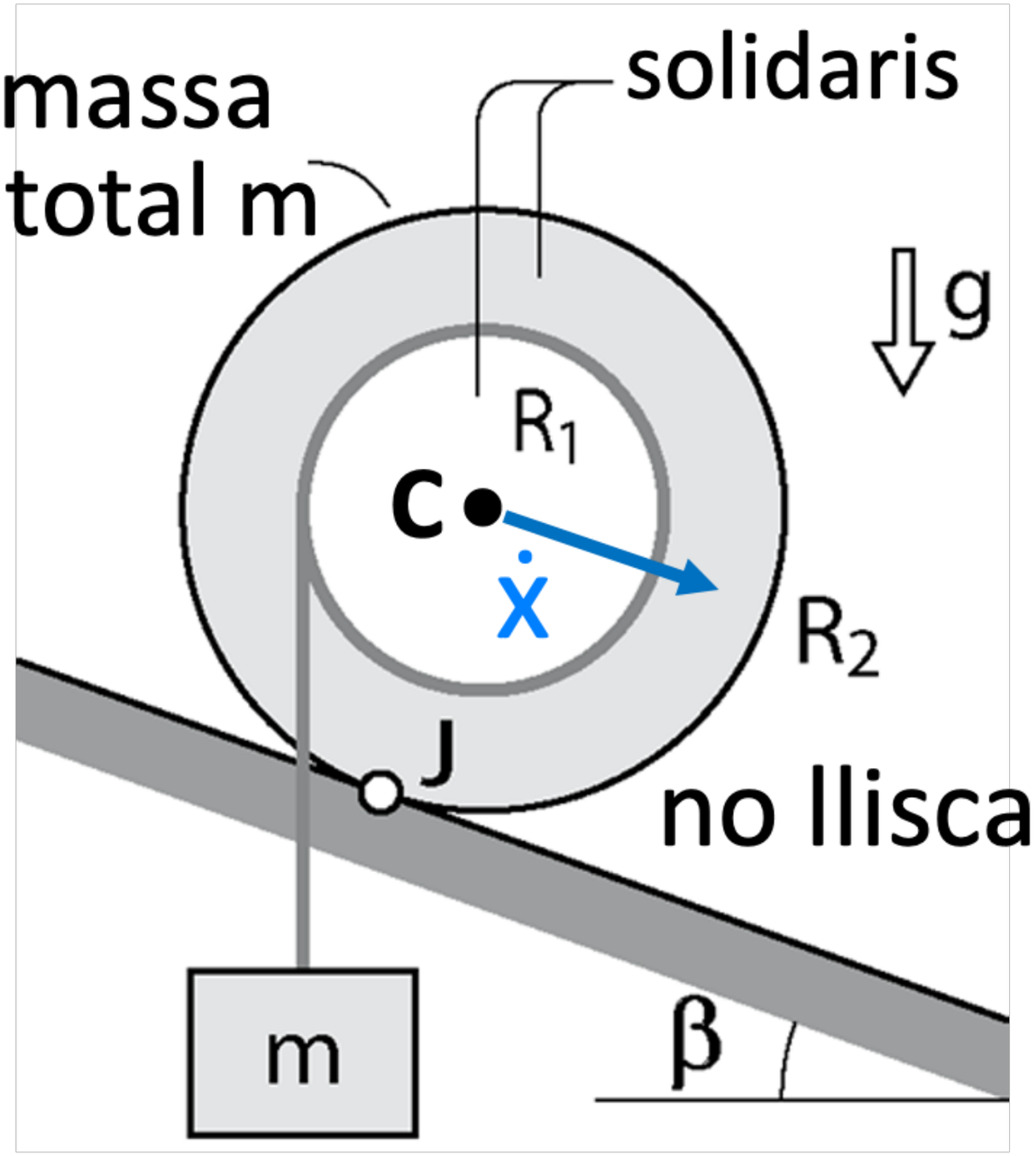
B = (1,2,3)

$$\boxed{\overline{H}_{RTG}(\mathbf{G})?}$$

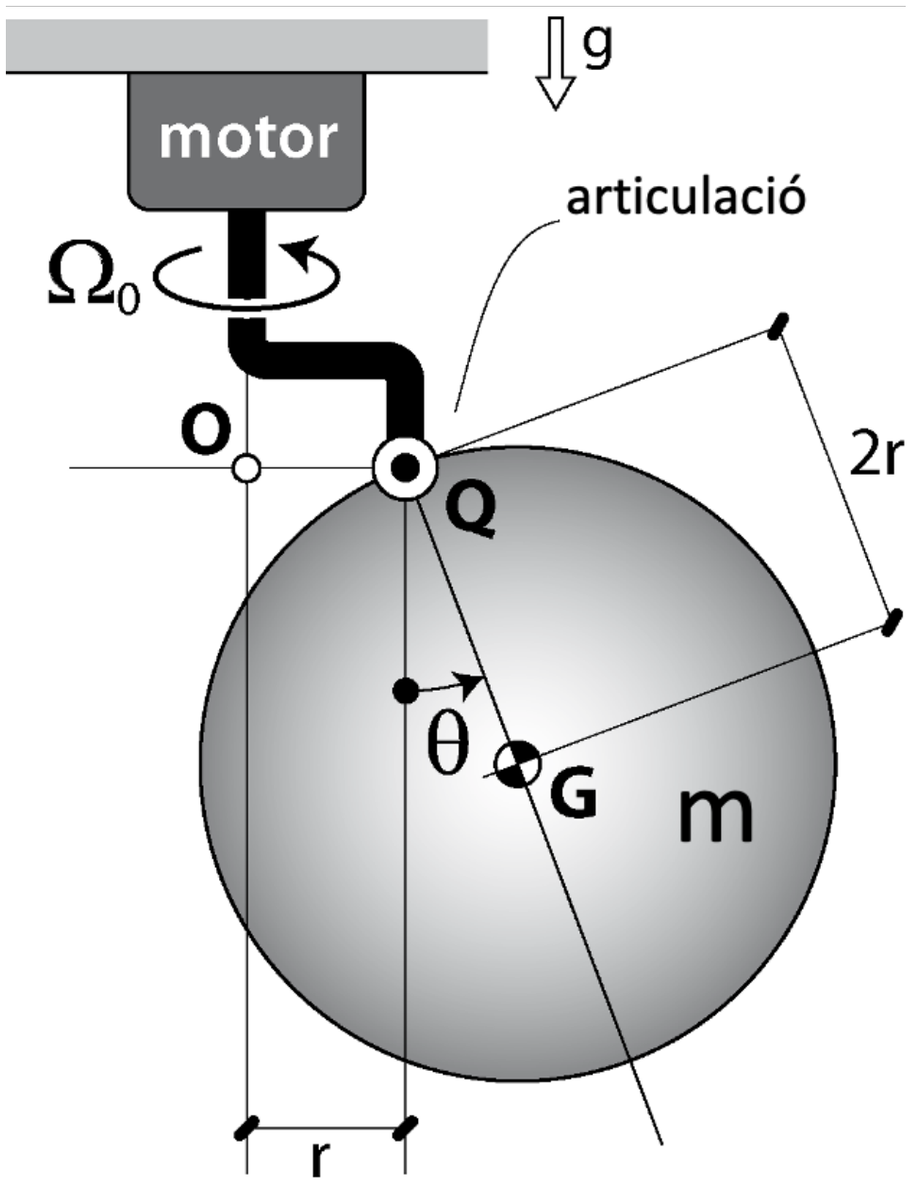


les rodes no llisquen

$$\boxed{\bar{H}_{RTC}(\mathbf{c})?}$$

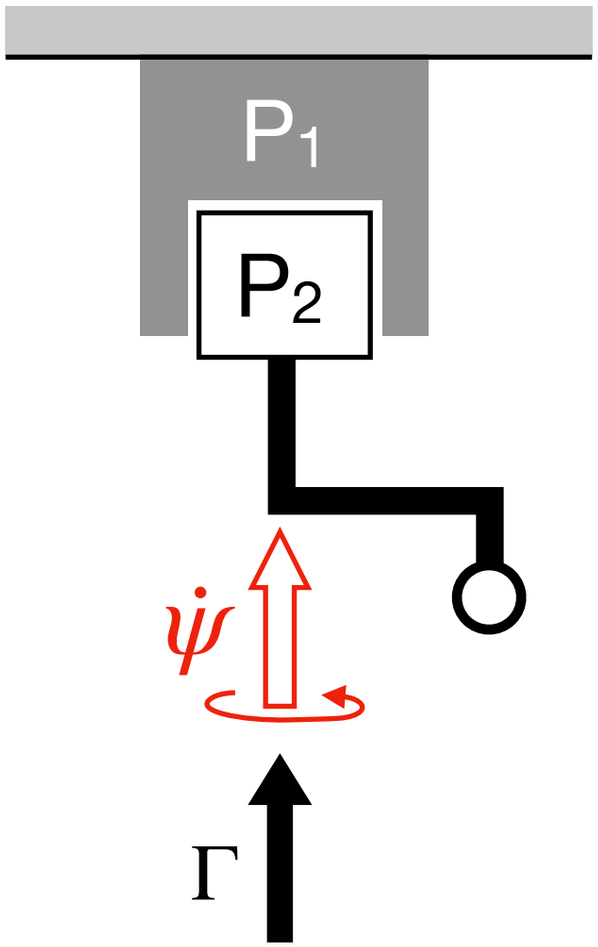


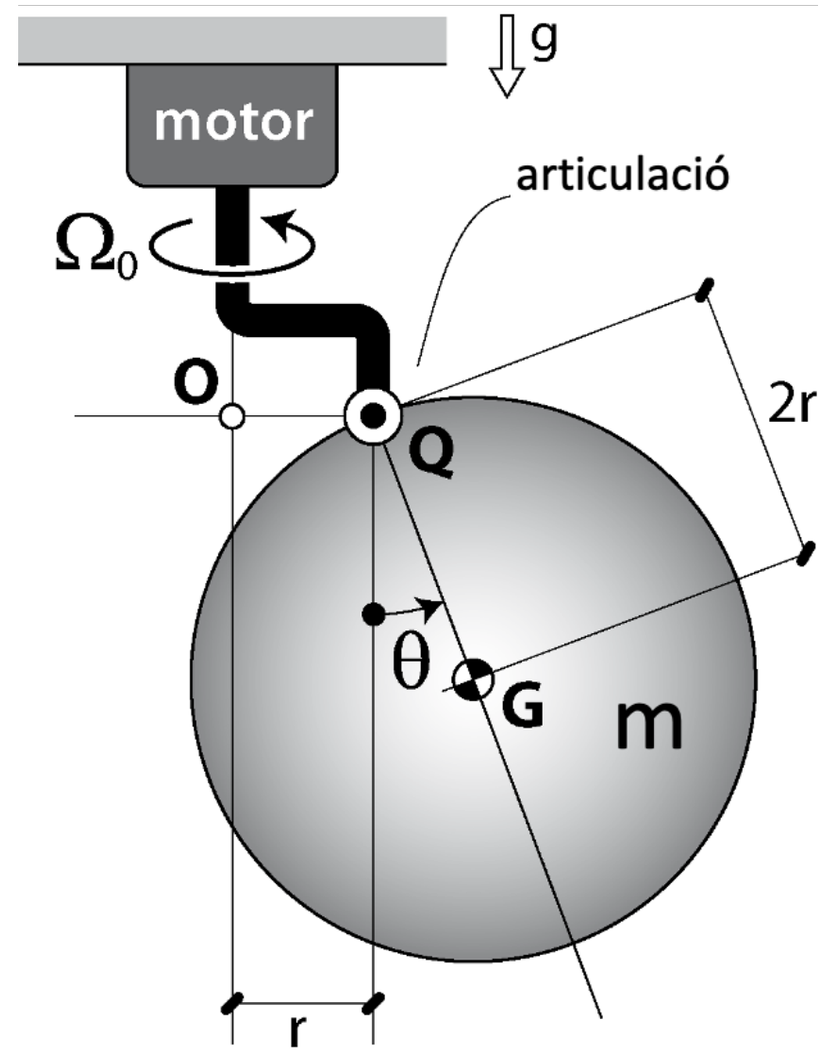
Pèndol esfèric giratori



- Eq. mov. per a θ ?
- Parell motor que garanteix Ω_0?

Tractament de motors





En general

En un motor:

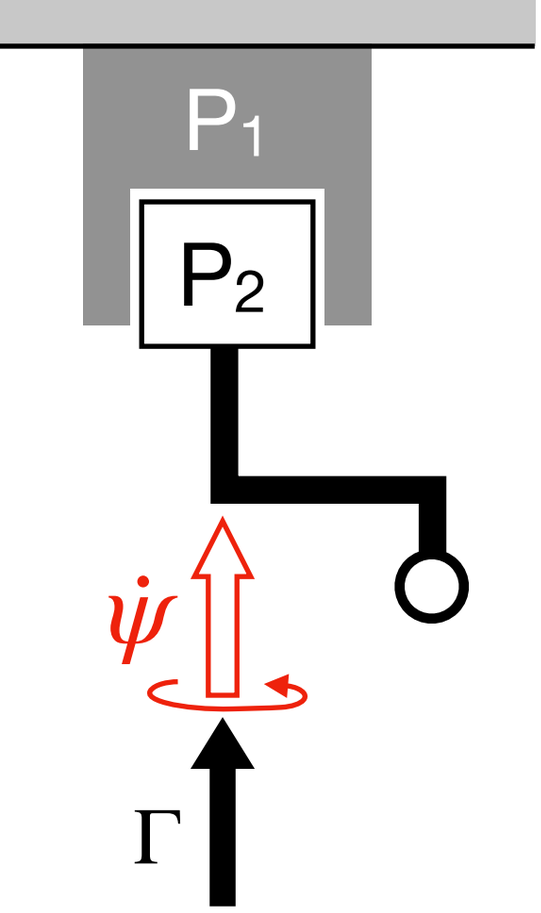
O bé sabrem Γ, i $\ddot{\psi}$ serà incògnita

O bé sabrem $\ddot{\psi}$, i Γ serà incògnita

En aquest exercici

$\dot{\psi} = \Omega_0 \;\; \Rightarrow \;\; \ddot{\psi} = 0$ (coneguda)

Γ serà incògnita



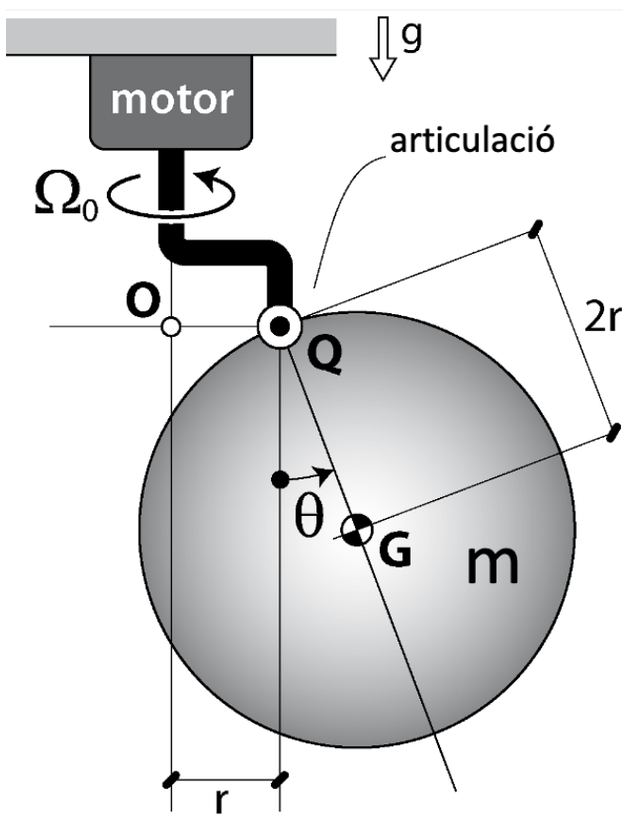
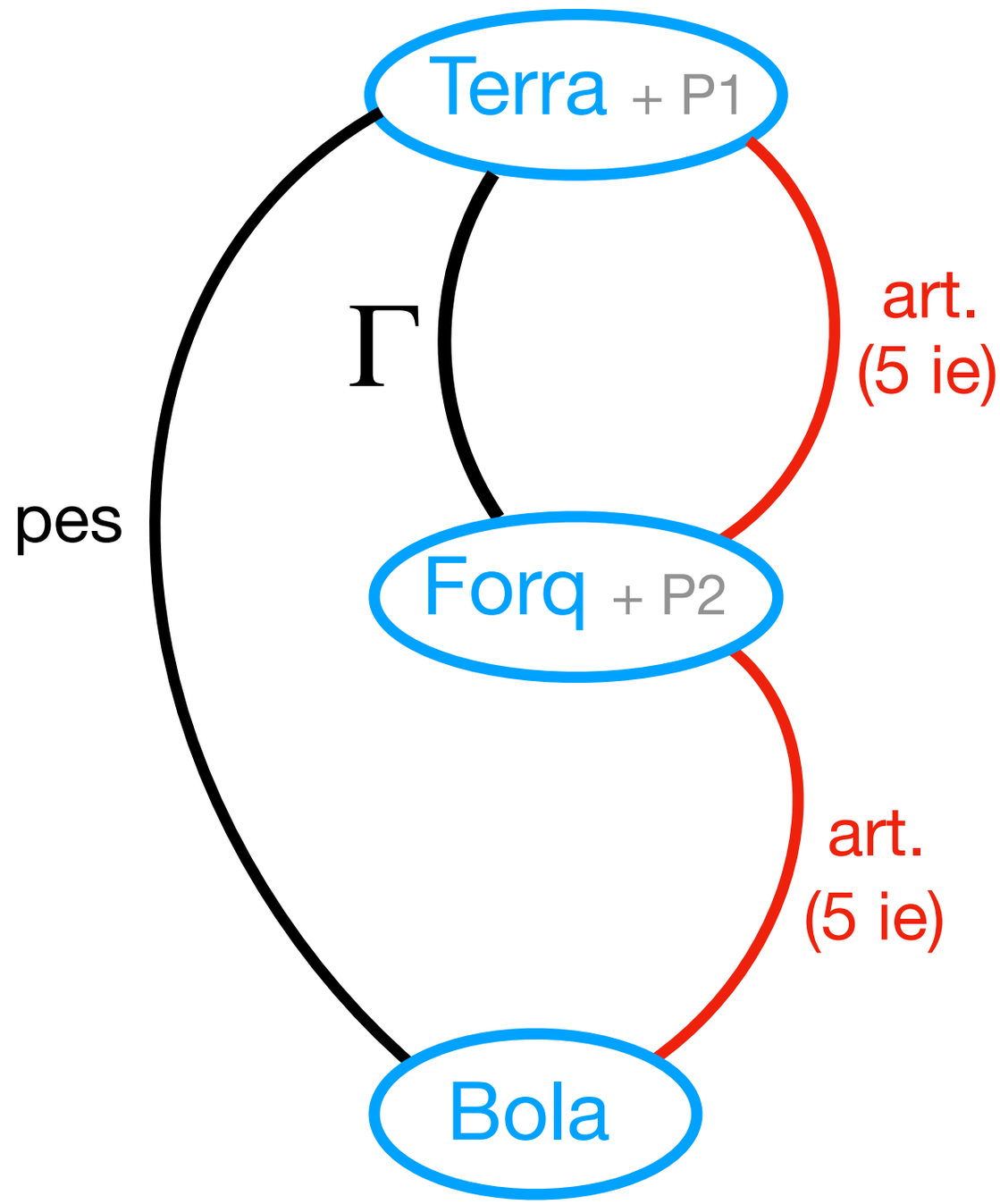
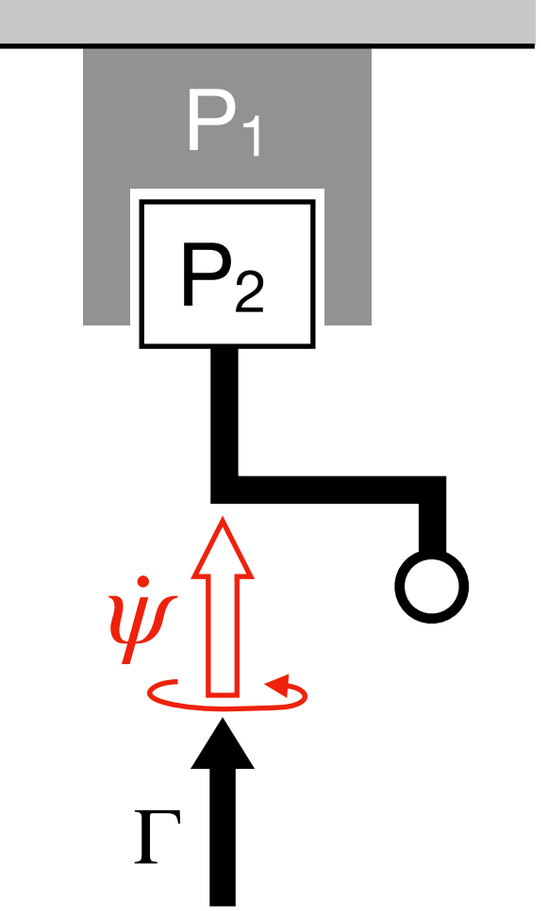
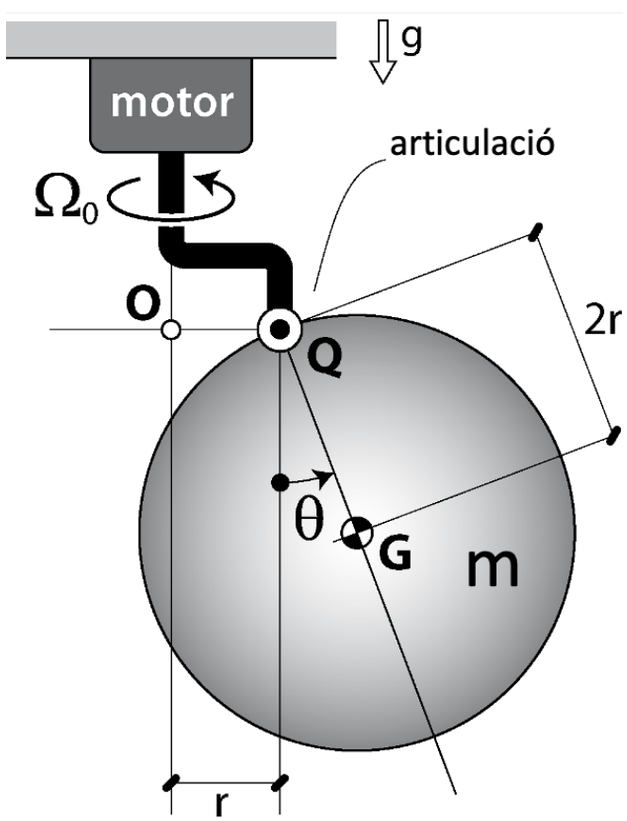


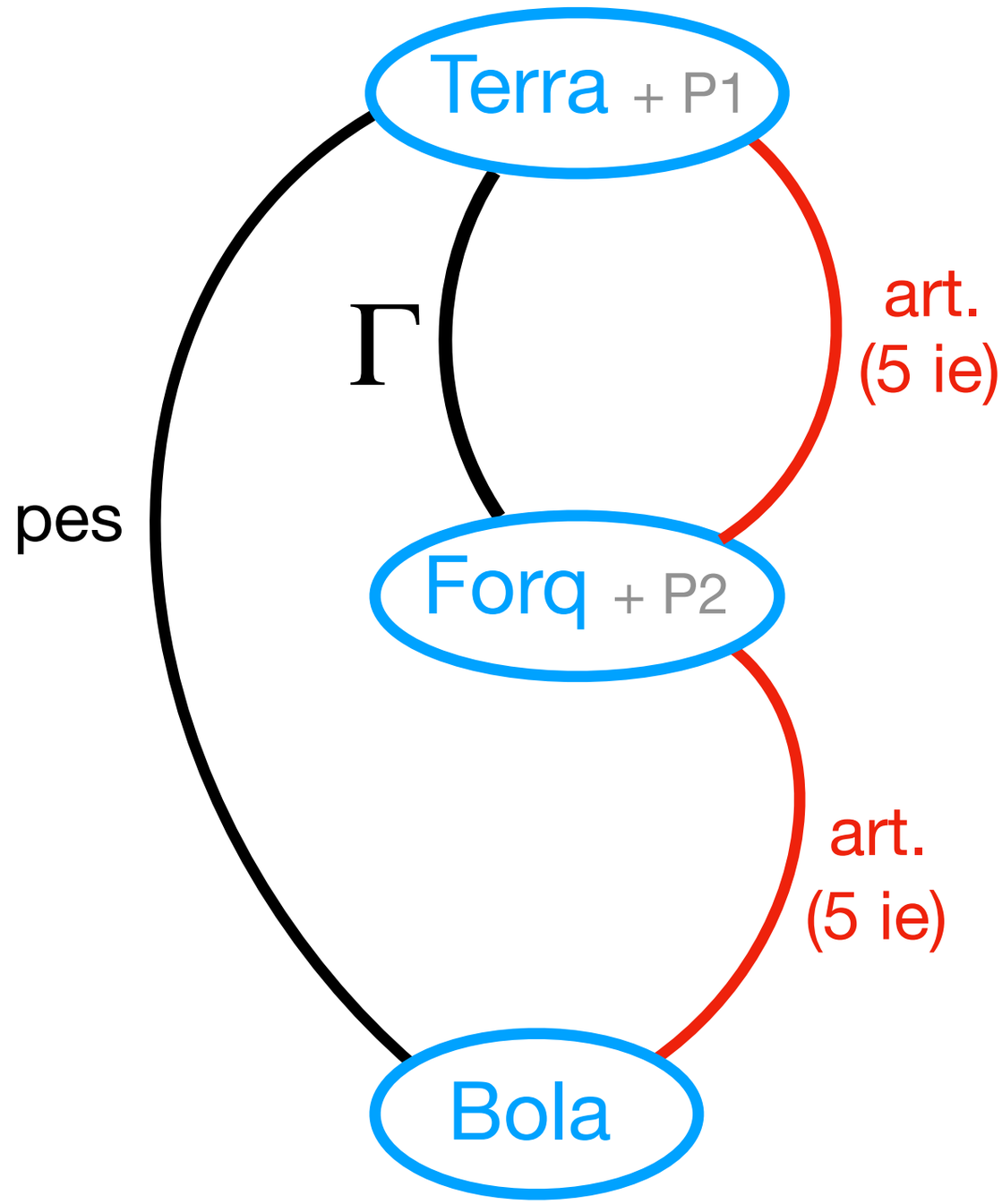
Diagrama general d'interaccions



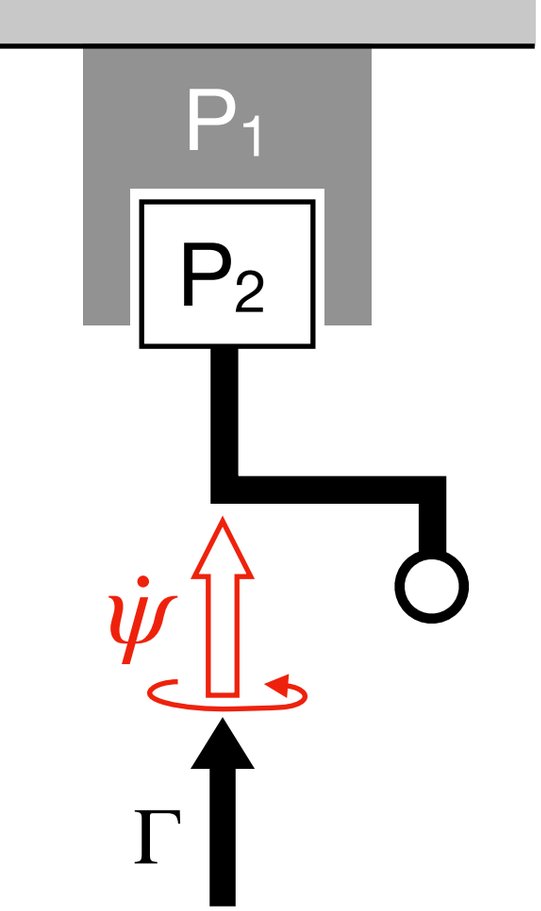
Problema dinàmic global

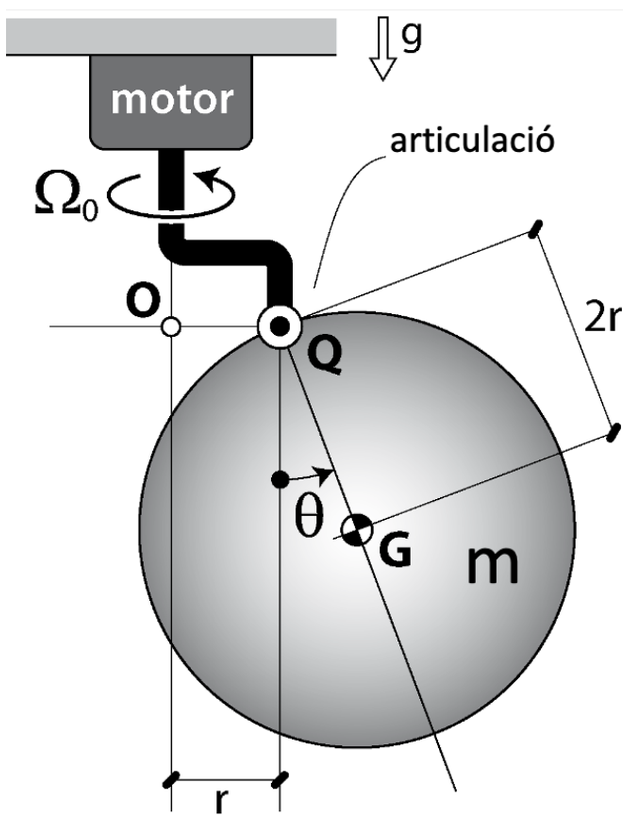


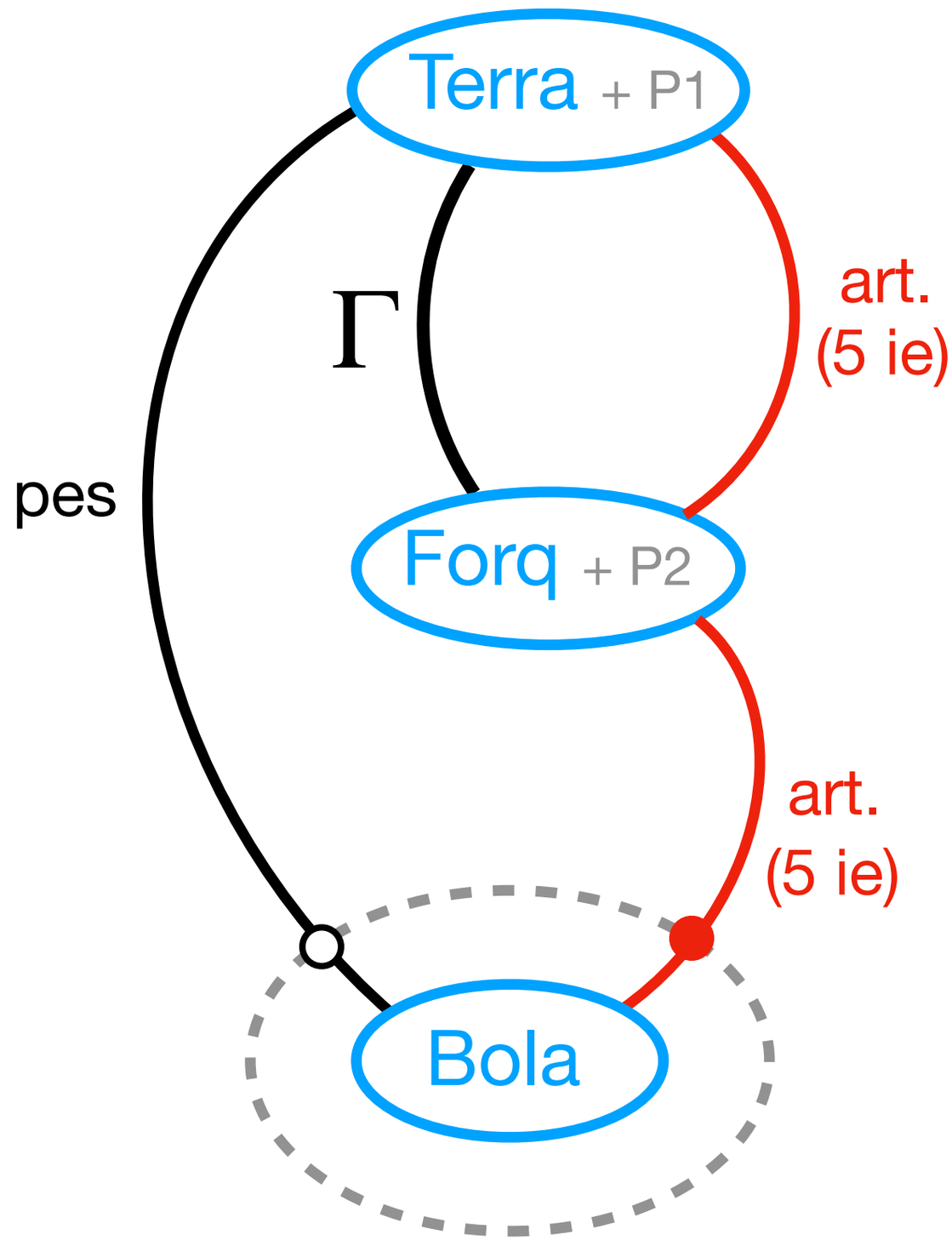




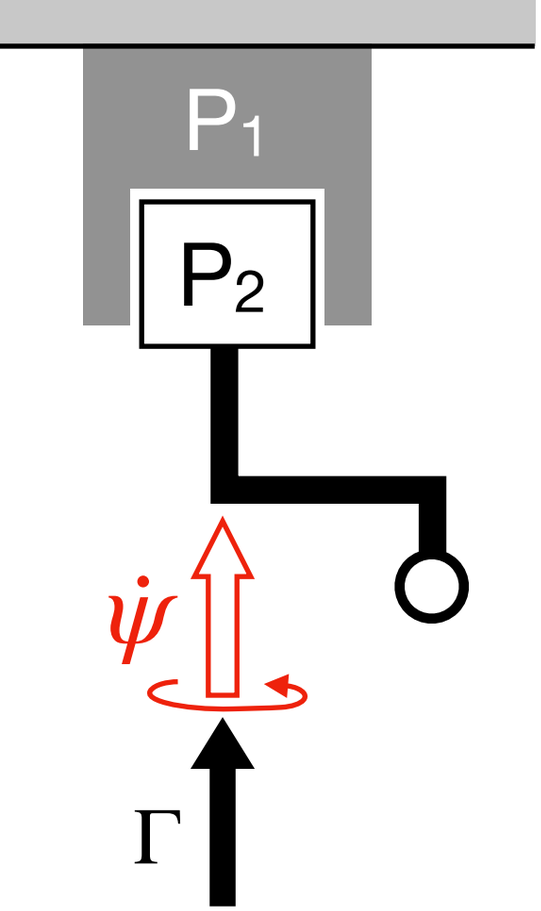
Sistema = Bola

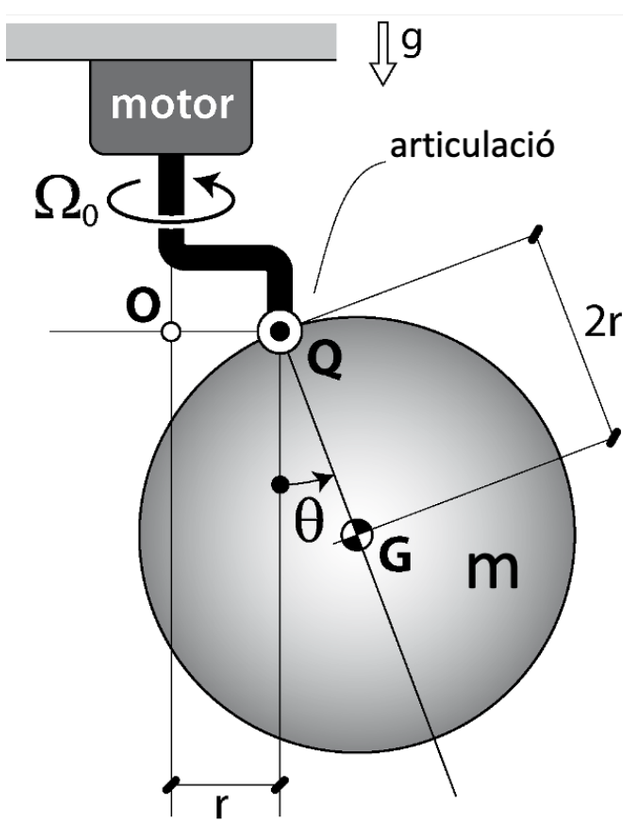


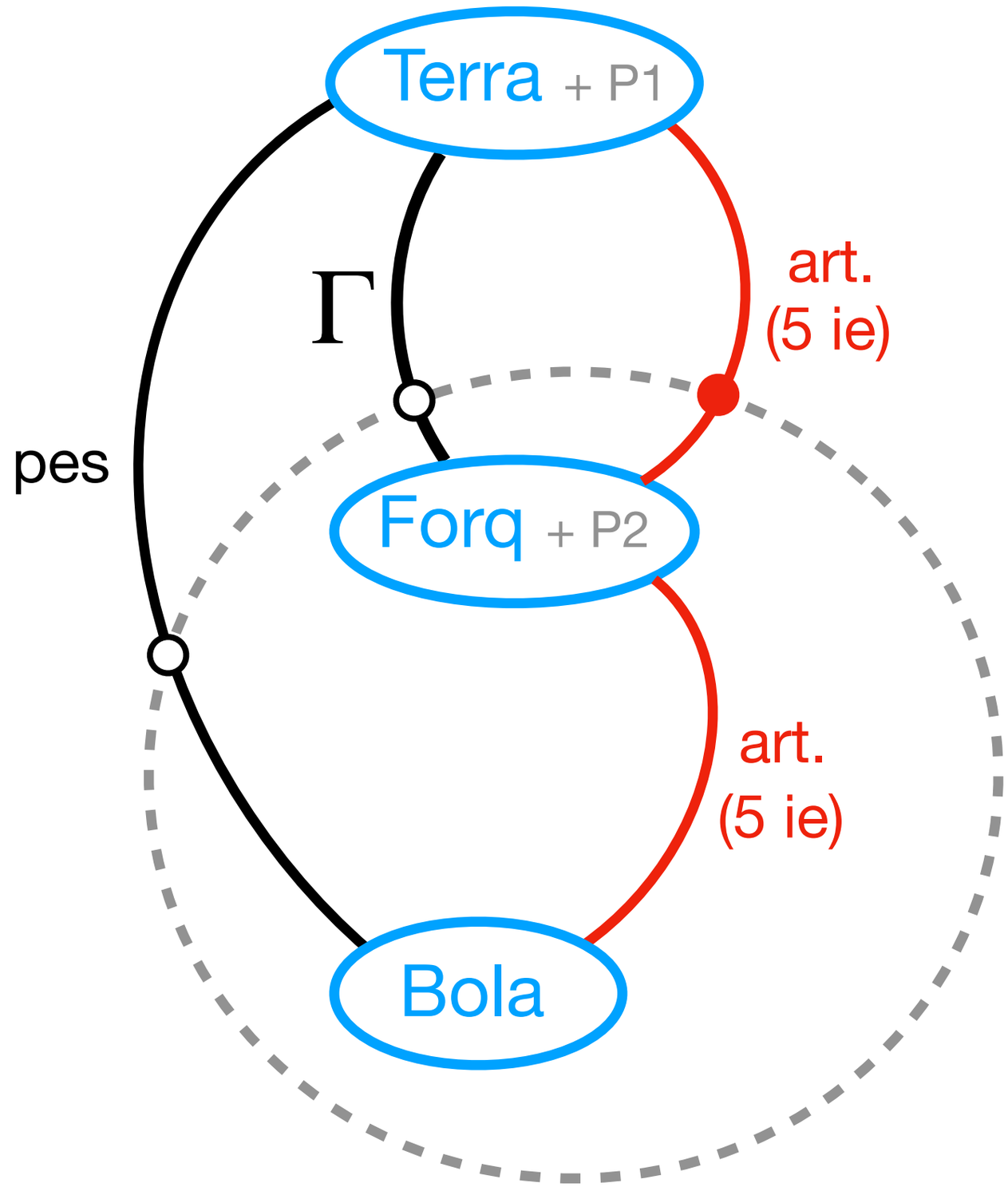




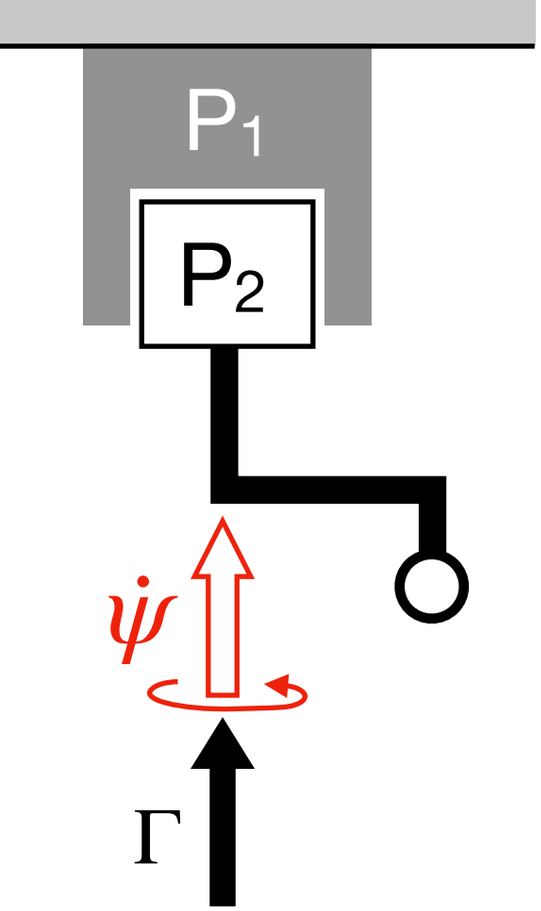
Sistema = Bola + Forq

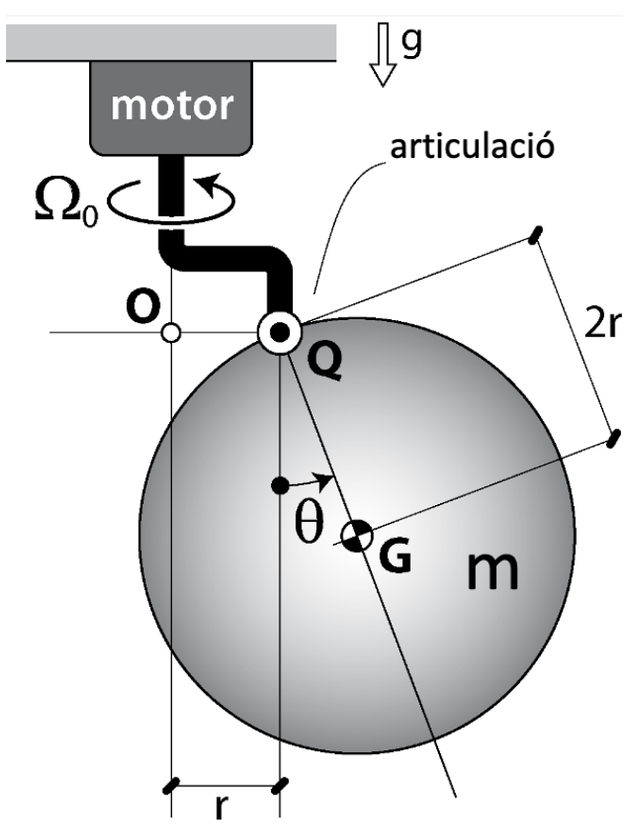


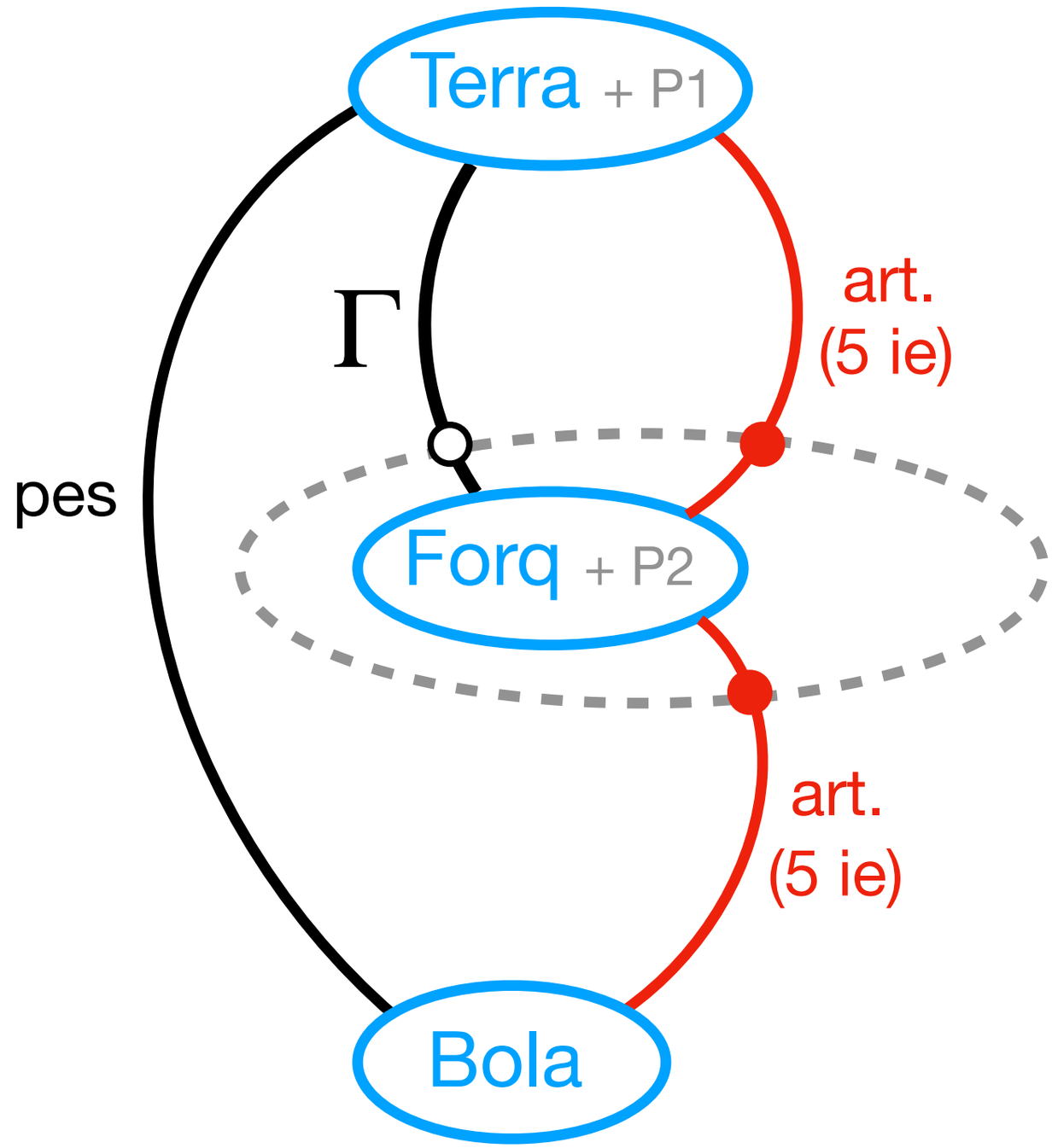




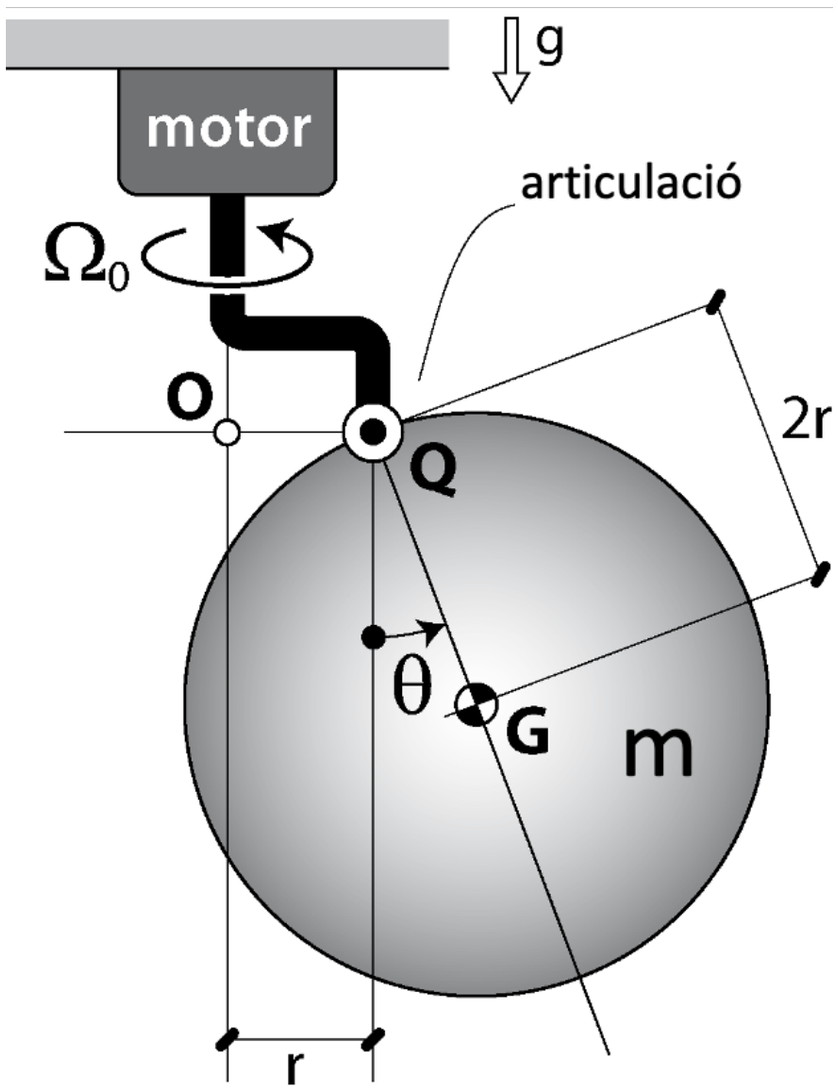
Sistema = Forq







Taula resum

Sistema	Incògn.	#incògn.	Problema
Bola	5 ie, $\ddot{\theta}$	6	**DET**
Bola + forq	5 ie, $\ddot{\theta}, \Gamma$	7	**INDET**
Forq	10 ie, Γ	11	**INDET**

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